

Matthew Choy

☎ (416) 371-9828 [in linkedin.com/in/matthewwychoy](https://www.linkedin.com/in/matthewwychoy) ✉ mchoy@uwaterloo.ca

SKILLS

CAD & Design: SolidWorks, Fusion360, AutoCAD, GD&T, DFMA, Machining, 3D Printing

Software: C++, Java, Git, FactoryTalk, Studio5000

Hardware: Circuits, Soldering, RC Systems, Computer Assembly, PLC, HMI, Electrical Drawings

WORK EXPERIENCE

Control Systems Co-op | *Lineside Industrial Automation* September 2026 – Present

- Developed and implemented PLC control logic using Studio 5000 and PCwin using Allen Bradley and Toyopuc PLCs, translating electrical and control requirements into reliable automation sequences for machine functionality.
- Designed and configured HMI interfaces using FactoryTalk View Studio ME, creating operator displays, controls, alarms, and status indicators to provide clear monitoring and control of automated systems.
- Mapped electrical drawings to PLC I/O, control logic, and HMI displays, ensuring accurate integration between field devices, PLC programs, and operator interfaces while troubleshooting discrepancies during system development.

Machine Vision Co-op | *Taymer International Inc.* January 2026 – April 2026

- Engineered and optimized 30+ UV and LED lighting configurations to generate high-quality images that enhanced machine vision model performance, leading to faster inspection times and more reliable quality control for clients.
- Designed and executed validation tests on machine vision inspection technology, identifying edge cases and improving detection accuracy and reliability across varying product conditions.
- Performed soldering and mechanical assembly of machine components based on professional engineering drawings, ensuring accurate integration of electrical and mechanical systems

ENGINEERING PROJECTS

Custom Computer Numerical Control (CNC) Machine December 2025 – Present

- Designed and built a compact 3-axis CNC machine using aluminum extrusions, linear rails, and lead screw drive systems, optimizing structural rigidity and axis alignment to improve positional accuracy.
- Implemented fully autonomous motion control using GRBL firmware on an Arduino, wiring and configuring stepper drivers, calibrating steps/mm, and tuning acceleration and feed rates for smooth, accurate multi-axis movement.
- Designed custom parts in SolidWorks and generated toolpaths, incorporating DFM principles such as tolerancing, geometry optimization, and designing for easier manufacturing and assembly.
- Reduced mechanical errors by iterating on anti-backlash solutions, bearing supports, and coupler alignment, significantly improving repeatability and positional accuracy under load.

VEX VSLAM Rover October 2025 – December 2025

- Built and programmed a VEX mapping rover that performs 360° scans and generates real time 2D maps using the data from the distance sensor.
- Developed C++ software to track robot position relative to an origin using motor encoders and inertial sensors, converting the data into a 128 x 128 on-screen boolean map.
- Designed a cable telescopic linear slider mechanism for adjustable scan heights and improved mechanical reliability.

RC Airplane June 2025 – August 2025

- Designed and built a RC Airplane, integrating a brushless motor, ESC, and LiPo battery system to meet thrust and flight requirements while achieving stability and an optimal centre of gravity.
- Built and soldered electrical systems, maintaining stable power and accurate signal transmission between the motors and control systems.
- Conducted 20+ iterative test flights to identify stability and control issues, refining mechanical and electrical configurations to improve overall flight performance.

EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering

September 2025 – Present